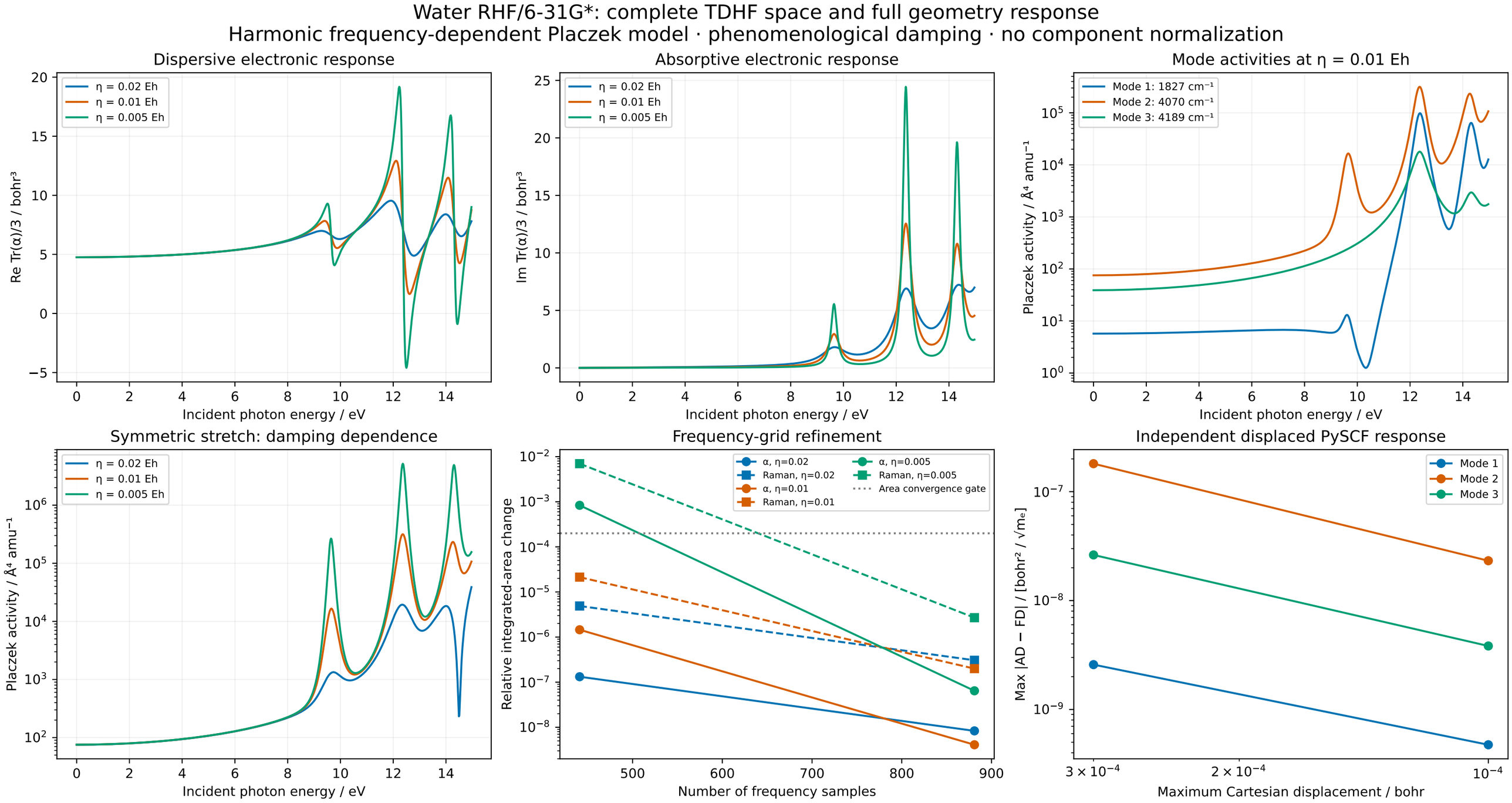
Water RHF/6-31G*: complete TDHF space and full geometry response
Harmonic frequency-dependent Placzek model · phenomenological damping · no component normalization
Dispersive electronic response
η = 0.02 Eh
η = 0.01 Eh
η = 0.005 Eh
Re Tr(α)/3 / bohr³
Incident photon energy / eV
Absorptive electronic response
Im Tr(α)/3 / bohr³
Mode activities at η = 0.01 Eh
Mode 1: 1827 cm⁻¹
Mode 2: 4070 cm⁻¹
Mode 3: 4189 cm⁻¹
Placzek activity / Å⁴ amu⁻¹
Symmetric stretch: damping dependence
Frequency-grid refinement
α, η=0.02
Raman, η=0.02
α, η=0.01
Raman, η=0.01
α, η=0.005
Raman, η=0.005
Area convergence gate
Relative integrated-area change
Number of frequency samples
Independent displaced PySCF response
Mode 1
Mode 2
Mode 3
Max |AD − FD| / [bohr² / √mₑ]
Maximum Cartesian displacement / bohr